



Research Paper

Weather, credit, and economic fluctuations: Evidence from China[☆]Zhenzhu Chen^a, Li Li^{b,*}, Yao Tang^{a,*}^a Department of Applied Economics, Guanghua School of Management, Peking University, 5 Yiheyuan Road, Haidian District, Beijing 100871, China^b School of Finance, Nankai University, 38 Tongyan Road, Haihe Education Park, Tianjin 300350, China

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ABSTRACT

We constructed an Extreme Weather Index to measure extreme weather risks in China. Analyzing macroeconomic data through a structural vector auto-regression model suggests that a negative weather shock leads to persistently low GDP and credit obtained by non-financial firms. In our regression analysis of a panel of firms listed in China, the negative effects of weather shocks on firm level loans were statistically and practically significant. Further analysis suggests that credit risk and expectations are two important impact channels. A high existing credit risk or low confidence among firm managers, amplifies the negative effects of extreme weather on loans.

1. Introduction

A large body of scientific literature indicates that climate change has led to more frequent extreme weather events, such as abnormal temperatures and precipitation (Stott, 2016; Allan et al., 2021). The years 2000 to 2019 saw a worldwide record of more than 11,000 extreme weather events, resulting in a total loss of \$2.56 trillion (Eckstein et al., 2021). Existing evidence suggests that adverse climate events significantly and immediately impact aggregate economic output, (Dell et al., 2012; Burke et al., 2015; Kim et al., 2021) and organizational performance (Carleton and Hsiang, 2016; Cai et al., 2018; Somanathan et al., 2021), through physical damage, labor productivity loss, and other channels. China, a large developing country and an important stakeholder in the global climate actions, is subject to many extreme weather events. Based on the Global Climate Risk Index constructed by Eckstein et al. (2021), China ranked 41st in climate risks from 2000 to 2019. But in terms of economic losses and fatalities, China ranked the first and fifth, respectively.

This study examines the impact of extreme weather events on the Chinese economy at both the macroeconomic level and firm level. Using daily data published by the China Meteorological Data Service Center, we constructed an Extreme Weather Index (EWI)

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to measure climate-related extreme weather risks in China. At the macroeconomic level, a structural vector auto-regression (SVAR) model is estimated for EWI, gross domestic product (GDP), consumer price index (CPI), interest rate, and total credit obtained by non-financial firms, from 2001 Q1 to 2020 Q4. Our microeconomic analysis included panel regressions of loans obtained, and value-added produced by listed firms on the EWI, from the 2006 Q1 to 2020 Q4.

We obtained three key findings at the macroeconomic level. First, extreme weather shocks have a significant negative impact on the GDP and credit financing. After a negative extreme weather shock hits the economy, the GDP and credit obtained by non-financial firms remain persistently low up to the 24th quarter. In particular, following a one standard deviation shock to EWI, GDP drops by 0.5% in the third quarter. Second, variance decomposition indicates that extreme weather shocks account for 14.2% of the variation in GDP in the long term, illustrating the importance of extreme weather risks to China's macroeconomy. Third, our counterfactual analysis indicates that if the central bank does not react to extreme weather shocks, the size of the drop in GDP following a one-standard-deviation extreme weather shock will worsen from -0.42% to -0.49% in the long term. Aggregate evidence confirms the contractionary effect of extreme weather shocks, and suggests that credit is an important mechanism for propagating the impact of extreme weather shocks on the real economy.

Our panel regression analysis which focuses on the significance of credit and credit-related factors in economic responses to extreme weather shocks, yields three principal findings. First, extreme weather events reduce loans obtained by listed firms and the drop in loans obtained precedes the drop in firm-level output. Second, we examine whether credit risk is an important channel of extreme weather affecting credit. We find supportive evidence in that expected default risk rises after extreme weather shocks. Consistent with the notion that collateral assets reduce credit risk, firms with higher collateral assets experience a smaller drop in credit, following extreme weather shocks. Macroeconomic factors that contribute to credit risk, such as tight monetary policies or economic uncertainty, worsen the negative effects of extreme weather. Third, expectations are another channel of impact. We find that extreme weather shocks cause stock analysts to downgrade the ratings of listed firms and reduce their profit forecasts. In addition, we examine two macro-level indicators related to general expectations, the entrepreneurs' confidence index and the production managers' index (PMI), and show that extreme weather shocks reduce loans by a larger magnitude when expectations are weak. Value-added of firms drop after extreme weather shocks, corroborating the macroeconomic finding that GDP is negatively affected by extreme weathers. Overall, the firm-level regressions provide strong evidence that credit risk and expectations are two important channels of extreme weather that affect credit.

Our research is related to the literature on the economic impact of extreme weather events. Researchers have studied and established the effects of temperature and other weather shocks, on GDP, employment, agricultural output, productivity, investment, health, and consumption (Dell et al., 2012; Graff Zivin and Neidell, 2014; Burke et al., 2015; Burke and Emerick, 2016; Zhang et al., 2017; Jessoe et al., 2018; Lai et al., 2022). The present study is closely related to Gallic and Vermandel (2020) and Kim et al. (2021), who estimated the impulse responses of economic variables to weather shocks using time-series data. The former study applied the SVAR model to data from New Zealand, and built general equilibrium model with an agricultural sector to explain the mechanism. The latter study used US data to estimate the responses of industrial output, CPI, unemployment rate, and federal funding rate, in a smooth transition vector auto-regression model (STVAR), discovering that the adverse effects of extreme weather becomes more prominent over time. While we also estimate the impact of weather shocks on output, our study differs from the above in that we emphasize the role of credit in the propagation of extreme weather shocks, and provide firm-level evidence to complement the macroeconomic analysis. In addition, we study the role of monetary policy in mitigating the adverse impacts of extreme weather events by counterfactual scenario analysis.

The present study also belongs to the emerging literature that assesses the financial implications of climate change (see Hong et al. (2020) for a review of recent research). Climate change elevates credit risk in sections of the financial market studied, but the effects are nuanced. In the bond market, climate risk causes high-exposure US counties to pay more underwriting fees and initial yields, to issue long-term municipal bonds (Painter, 2020). In the real estate market, climate risk negatively affects the prices of residential property (Giglio et al., 2021) (albeit the effect can be muted by weak beliefs about climate change (Baldauf et al., 2020)), induces risk sharing by mortgage securitization in hurricane prone regions (Ouazad and Kahn, 2019), but fails to capture the attention of mortgage lenders in property valuation (Garbarino and Guin, 2021). Relative to the literature, we choose to focus on the effect of climate change on credit access of firms in the context of China.¹ This new focus is practically important because, as typical in developing economies, the Chinese financial system is dominated by banks and credit plays the most important role in the Chinese firms' financing.

The remainder of this paper is organized as follows. Section 2 describes the construction of the EWI. The results of the macroeconomic and firm-level analyses are reported and discussed in Sections 3 and 4, respectively. Section 5 concludes the paper and discusses future research.

2. Measuring extreme weather in China

Our construction of EWI follows the methodology of Actuaries Climate Index (ACI), which was proposed by the American Academy of Actuaries, Canadian Institute of Actuaries, and Casualty Actuarial Society and Society of Actuaries in 2020. As the

¹ By employing flood and wildfire risks as proxies for climate risk, Meisenzahl (2023) estimate that a one-standard-deviation increase in climate risk reduces county-level balances in banks' portfolios by up to 4.7 percent in counties with large loan balances between 2014 and 2020. In contrast to the bank perspective adopted by Meisenzahl (2023), we examine the effect of extreme weather on credit from the perspective of borrower firms.

ACI is constructed with information on high temperature, low temperature, precipitation, drought, wind, and sea level change, it essentially covers several dimensions of extreme weather. However, weather which typically refers to short term atmospheric conditions such as precipitation, temperature, humidity and wind speed, is different from climate which describes the average condition of the atmosphere over a long period of time. Therefore, we believe that it is more accurate to use the term “Extreme Weather Index”, instead of “Actuary Climate Index” in the current study. While the focus of the current study is on extreme weather, we certainly recognize that extreme weather events are closely linked to climate change. The Intergovernmental Panel on Climate Change (IPCC)’s Sixth Assessment Report released in 2021 noted that “the human-caused rise in greenhouse gases has increased the frequency and intensity of extreme weather events”.

The EWI index aims to monitor and quantify the frequency of extreme weather events, helping policymakers and the general public to understand the potential impact of climate change on the economy and society. Using the Chinese Surface Climate Daily Data, published by the China Meteorological Data Service Center, we applied the formula of ACI to construct the monthly time series of the Chinese EWI.

In the first step of constructing EWI, for each of the 840 surface stations which provide comprehensive coverage of weather in Mainland China, we retrieve the value of average temperature, precipitation, and wind speed of each calendar day. Based on these daily values, we construct the following monthly EWI component variables for each of the surface stations.

1. Extreme high temperature (P90): the variable takes a value of 1, if the average daily temperature is above the 90th centile of all daily averages in the sample, and 0 otherwise. The daily value can then be aggregated to the monthly level, which takes the value in [0,31].

2. Extreme low temperature (P10): the variable takes a value of 1, if the average daily temperature is below the 10th centile of all daily averages in the sample, and 0 otherwise. We then aggregate the daily value to the monthly level.²

3. Concentrated precipitation (R): the variable is equal to the maximum precipitation level in a 5-day window within a month.

4. Drought (D): the maximum number of consecutive days with daily precipitation less than 1 mm.³

5. Strong wind (W): the variable takes a value of 1, if the daily average wind power is above the 90th centile of all daily averages in the sample, and 0 otherwise. We then aggregate the daily value to the monthly level.

In the second step, in order to aggregate the five component variables which carry different units, we convert them into standardized anomalies. For each component variable, the standardized anomalies are calculated by taking the difference between the value of the component variable in a month and its mean value in the reference period, and dividing the difference by the standard deviation of the component variable in the reference period. Thus, for each of the five extreme-weather component variables, the standardized anomaly in a month measures the degree of deviation of the monthly value from the mean the reference period for a surface station. For example, a component variable value of 2.0 indicates that the value is two standard deviations above the mean value of that index during the reference period.

As our main data range is from 1990 to 2020, we take the first half (January 1990 to December 2005) as the reference period. For each month $m = 1, 2, \dots, 12$ and for each component variable x , we subtract from x the mean of x values in month m of all years in the reference period, and divide the difference by the corresponding standard deviation of x from the reference period. For instance, the standardized anomaly $X_{i,t}^{std}$ of the component P90 (extreme high temperature) at station i and in August 2015 is calculated as $\frac{P90_{i,Aug\ 2015} - \mu_{P90,i}^{1990-2005}}{\sigma_{P90,i}^{1990-2005}}$, where $\mu_{P90,i}^{1990-2005}$ is the mean of P90 in station i in all August in the 1990–2005 period and $\sigma_{P90,i}^{1990-2005}$ the corresponding standard deviation. Clearly, the standardized variables measure extreme weather patterns relative to the reference period. As we compute $X_{i,t}^{std}$ month by month, the standardized variable naturally accounts for the seasonal pattern of the extreme-weather component variable at the monthly frequency.

In the last step of construction, the monthly sequences of standardized anomaly $X_{i,t}^{std}$ from 840 stations in the mainland China are then averaged across the stations. This yields five aggregate time series $P90_t^{std}$, $P10_t^{std}$, R_t^{std} , D_t^{std} , W_t^{std} . The monthly Chinese EWI is then defined as:

$$EWI_t = P90_t^{std} - P10_t^{std} + R_t^{std} + D_t^{std} + W_t^{std} \quad (1)$$

where superscript *std* denotes the standardized variables. We follow the original definition of the ACI, such that $P10_t^{std}$ enters with a negative sign because a reduction in cold extremes is posited to increase extreme weather risks. The mean EWI in the sample range was -0.073, with a standard deviation of 1.89. The variable ranged from -4.09 to 8.91.

Fig. 1 shows the resulting monthly Chinese EWI. The five-year average of the EWI initially trended downward, before rapidly rising after 2015. The end of the sampling period saw the highest value of 5-year average EWI, since 2000. In addition to the long-term trends, EWI substantially varies and indicates frequent short-term extreme weather shocks. The monthly EWI reached its maximum value in September 2019 because $P90_t^{std}$ and D_t^{std} were extremely high. Indeed, the China Meteorological Administration

² In the construction of Actuaries Climate Index (<https://actuariesclimateindex.org/data/component-definitions/>), the frequency of daily temperatures above 90th percentile or below the 10th percentile is also utilized to define extreme high temperatures or extreme low temperatures, respectively.

³ In American Academy of Actuaries’ method, Drought is defined as the maximum number of consecutive dry days (daily precipitation less than 1 mm) per year, and linear interpolation between two consecutive yearly values are used to approximate monthly and seasonal values. In the case, linear interpolation is adopted because only the count of the number of consecutive dry days per year is available in their raw datasets. In comparison, because daily precipitation for each station is available in Chinese Surface Climate Daily Data, we are able to directly count the maximum number of consecutive dry days at the station level for each month. There is no need to use linear interpolation.

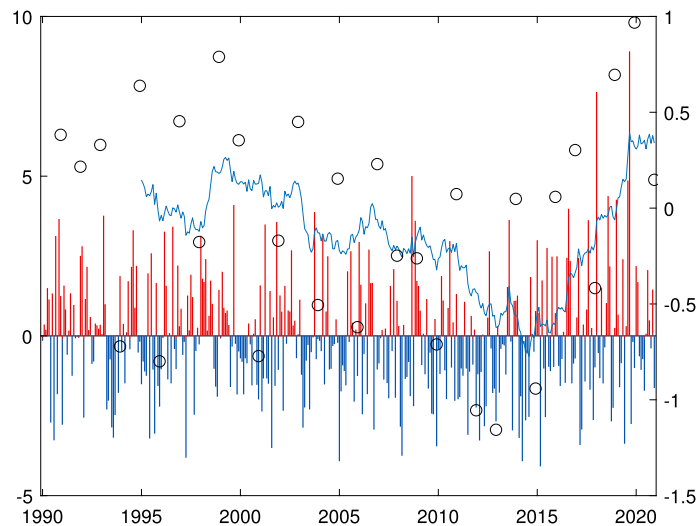


Fig. 1. Monthly Actuaries Climate Index (EWI) of China. Source: Author's calculations. The bars are the monthly EWI values (left axis), the black dots are the annual EWI values (right axis), and the solid blue line (right axis) is the five-year moving average of the EWI index. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

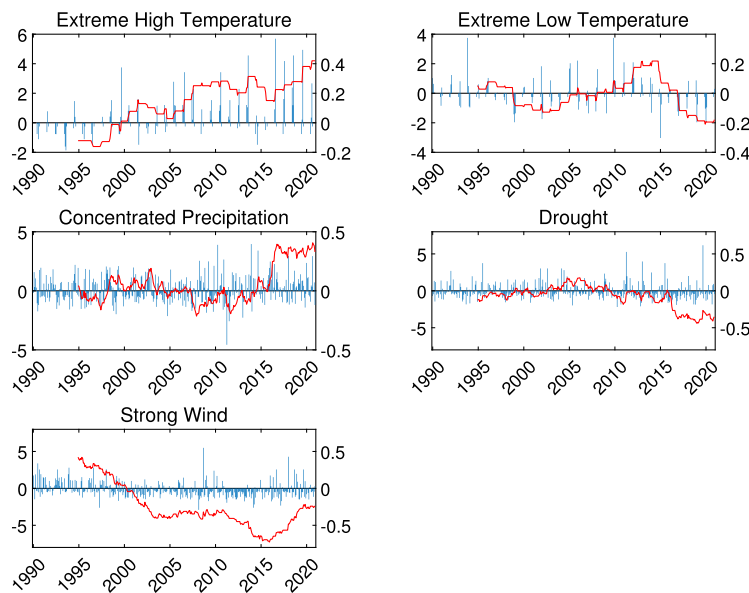


Fig. 2. Components of Monthly EWI of China. Source: Author's calculations. The blue bars are the monthly series for the five components of the EWI index, and the red lines are the corresponding 5-year moving averages.

reports experiences of a 50% to 90% decreased precipitation since late July, in the eastern area of Hubei Province, the central and eastern areas of Hunan Province, most of Jiangxi Province, the southern area of Anhui Province, and the central and northern areas of Fujian Province, relative to normal years. In the same month, precipitation level was lowest, relative to other Septembers since 1961. The national average temperature was 1.1 degrees Celsius higher than normal, reaching the third-highest level in Septembers since 1961.

We plot the five components of EWI from 1990 to 2020, in Fig. 2. Post 2015 saw frequent cases of extremely high temperatures, with rarer occurrence of extremely low temperatures, indicating a rise in average Chinese temperature. Incidences of concentrated precipitation reached the lowest levels in spring and summer of 2011. Increasingly frequent and severe droughts peaked in August 2019. The time series for strong winds, showed notable spikes in September 2008 and January 2018.

To validate the EWI, we compared our time series with the China Climate Risk Index (available from July 2016 to December 2018), published by the National Climate Center (NCC) of China, and found the correlation coefficient between the two series was 0.44. The left panel of Fig. 3 indicates the similarity between the two series, in major movements; however, the NCC China Climate Risk Index is much smoother than ours.

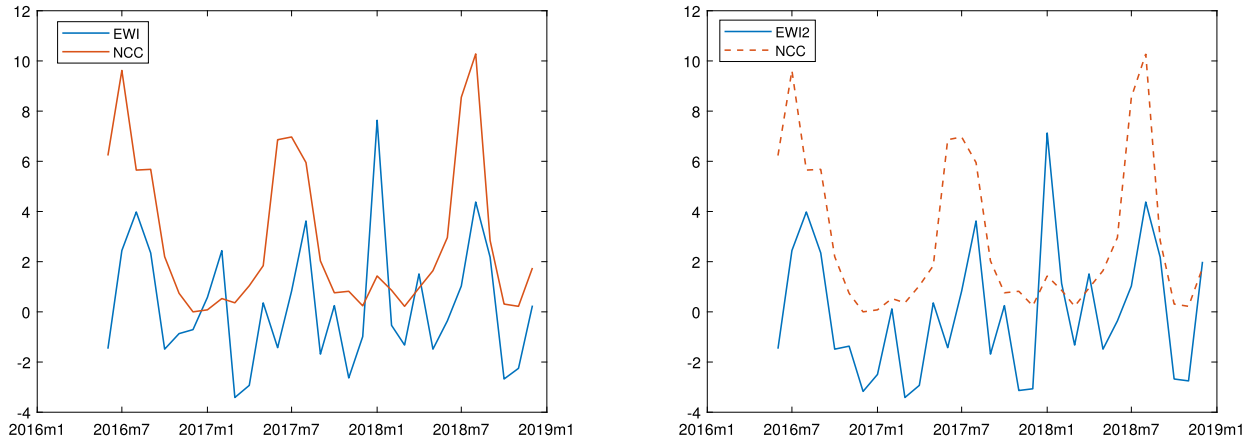


Fig. 3. EWI vs NCC China Climate Risk Index. Source: Authors' calculations and the National Climate Center (NCC) of China. The EWI in the left panel is constructed using the original definition. The EWI in the right panel adopts an alternative definition in which extreme colds are presumed to add to weather risks.

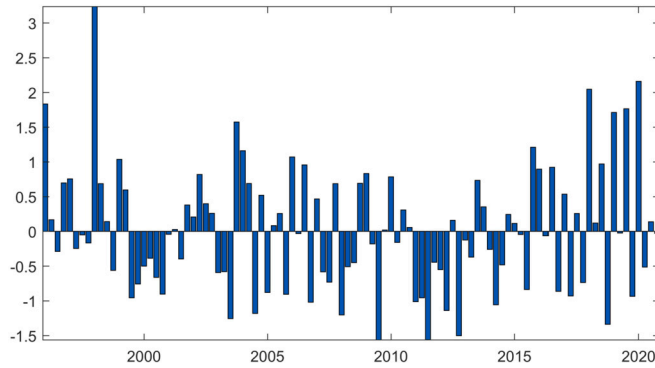


Fig. 4. Quarterly EWI of China. Source: Author's calculations. The bars are the quarterly EWI values which are obtained by averaging the monthly EWI index.

One difference in methodology is that the NCC index regards extreme cold as a factor positively related to extreme weather risks, whereas in our index (constructed by following the methodology of the American Academy of Actuaries and Canadian Institute of Actuaries), the relationship is presumed to be negative. Thus, we compute an alternative EWI series, in which the extreme cold variable ($P10^{std}$) enters with a positive sign. The correlation coefficient between the alternative EWI index and NCC China Climate Risk Index increases to 0.50. Alternative EWI is shown in the right-hand panel of Fig. 3. As shown in the robustness checks, our results are not sensitive to the treatment of extreme cold.

Overall, based on official Chinese meteorological data, our EWI series provides a reasonable measure of extreme weather risks from 1990 to 2020. To match with the quarterly macroeconomic and firm-level variables, we convert our monthly EWI into a quarterly EWI (plotted in Fig. 4) by taking the quarterly mean. The mean, minimum, maximum and standard deviation of the quarterly EWI are, -0.017, -1.66, 2.16, and 0.86.

3. Effect of extreme weather risks on the macroeconomy

To examine the effect of weather risks on China's macroeconomy, we estimate a quarterly five-variable SVAR of EWI, gross domestic product (GDP), consumer price index (CPI), interest rate (rate), and total credit obtained from non-financial firms (Credit). The SVAR is specified as

$$B_0 Y_t = c + B_1 Y_{t-1} + B_2 Y_{t-2} + \dots + B_p Y_{t-p} + w_t \quad (2)$$

where $Y_t = \{EWI, \ln(GDP), \ln(CPI), Rate, \ln(Credit)\}$ is the observation matrix, c is the intercept vector, and $B_0, B_1, \dots, B_p$ are the coefficient matrices. Vector w_t is the vector of structural shocks that satisfy $E(w_t) = 0$ and $E(w_t' w_t) = 0$.

As we are primarily interested in identifying extreme weather shocks, similar to Kim et al. (2021), we put the EWI in the first position in the SVAR, and perform a Cholesky decomposition to recover structural shock to EWI. The ordering of variables means that extreme weather shocks have contemporaneous effects on all other variables but not vice versa. We take an agnostic position on the temporal interactions between the other variables.

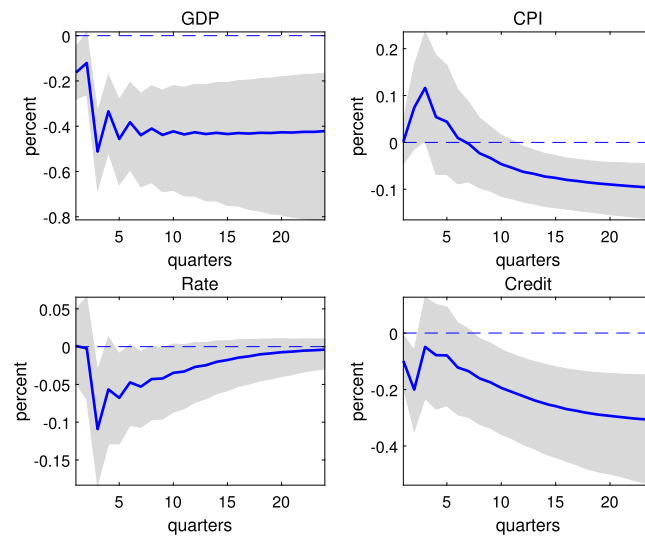


Fig. 5. Impulse responses to a one standard deviation extreme weather shock. Source: Authors' calculations. Note: The solid lines represent the mean impulse responses and the shaded area represents the 95% confidence interval.

We obtain the GDP, CPI, and interest rate from the WIND database. Our measure of interest rate is the 7-day inter-bank repo rate (the R007 interest rate), which is a key money market interest rate in China. We choose R007 over similar interest rates because it is available for a long period and because it is a market-based measure of the cost of funds. The total credit obtained from non-financial firm leverage was retrieved from the Research Center for the National Balance Sheet. We set the lags to two, based on the Akaike information criterion (AIC), Bayesian information criterion (BIC) and Schwarz criterion (SC). Our sample spans from 2001Q1 to 2020Q4.

The estimation of the SVAR model yields the impulse response functions (IRFs) in Fig. 5. From the figure, it can be seen that after a one-standard-deviation extreme weather shock, GDP displays a persistent negative response, with a drop of 0.2% in the first quarter and a maximum drop of 0.5% in the third quarter. The subsequent responses are smaller in magnitude; however, the drop in GDP remains approximately 0.4%. The CPI inflation response is hump-shaped, and while the response is not initially significant, it becomes negative and significant starting from the 12th quarter and reaches approximately -0.1%. The results indicate that extreme weather shocks leading to persistently low economic activities.

The negative response of the interest rate is short-lived, as it drops after two quarters, reaching the lowest point of -0.1% in the third quarter, and returning to zero in the long-term. The initial IRFs of interest are similar to those of GDP, indicating that the central bank is likely to ease monetary policy in response to extreme weather shocks. However, such adjustments in monetary policy are temporary and do not fully counteract the long-term negative impact of extreme weather shocks.

Fig. 5 shows that the credit obtained by non-financial firms drops at the impact. Despite the second quarter experiencing slightly improved credit access (interest rate is lowest here), it exhibits a long-term significant decline. This pattern again suggests that short-term monetary easing is insufficient for averting the long-term impact of extreme weather shocks on credit access.

To quantify how the impact of extreme weather risks changed over time, we performed historical and variance decomposition for the estimated SVAR. First, after detrending and demeaning the variables in SVAR, we decompose the *actual* deviation of a variable from its trend into contributions from external shocks to the five variables in each period. The visual results are shown in Fig. 6.

The impact of extreme weather risks on the GDP varies over time. A visual inspection of Fig. 2 and Fig. 6 suggests that between 2006 and 2011, the adverse effect of weather shocks is likely caused by the fact that the extreme high temperature indicator ($P90^{std}$) is above sample average. Between 2011 and 2015, weather shocks had either an insignificant or a positive effect on GDP growth. During this period, the indicator for concentrated precipitation (R^{std}) was historically low, and the extreme low temperature indicator ($P10^{std}$) became more positive. From 2013 to 2015, the strong wind indicator (W^{std}), drought indicator (D^{std}), and concentrated precipitation indicator (R^{std}) were all subdued.

In 2020, the extreme weather shocks increased significantly, explaining a 1% drop in GDP. Notable extreme weather events in 2020 were a lower number of occurrences of extremely low temperatures and more cases of concentrated precipitation during the first quarter. Meanwhile, our SVAR model implies that other shock, such as shock to GDP during the COVID-19 pandemic is the main reason behind the collapse in the growth rate, as it accounts for a 6% drop in GDP.

Second, we perform a variance decomposition for the estimated SVAR to assess the importance of different shocks in explaining the fluctuations of the variables. Table 1 presents the variance decomposition in the 1st, 4th, 8th, 16th, and 200th quarters, among which the 200th quarter approximates the infinite horizon. The impact of extreme weather risks remains significant over the long term. Taking GDP as an example: extreme weather risks accounts for only 2.8% of the variance in GDP Q1, but in the Q16, this fraction increases to a maximum of 18.3%. Over time, it settles to the long-term level of 14.2% in the 200th quarter. Similarly, extreme weather shocks negligibly affect the variation of CPI in Q1. However, they account for 11.5% of the long-term CPI variation,

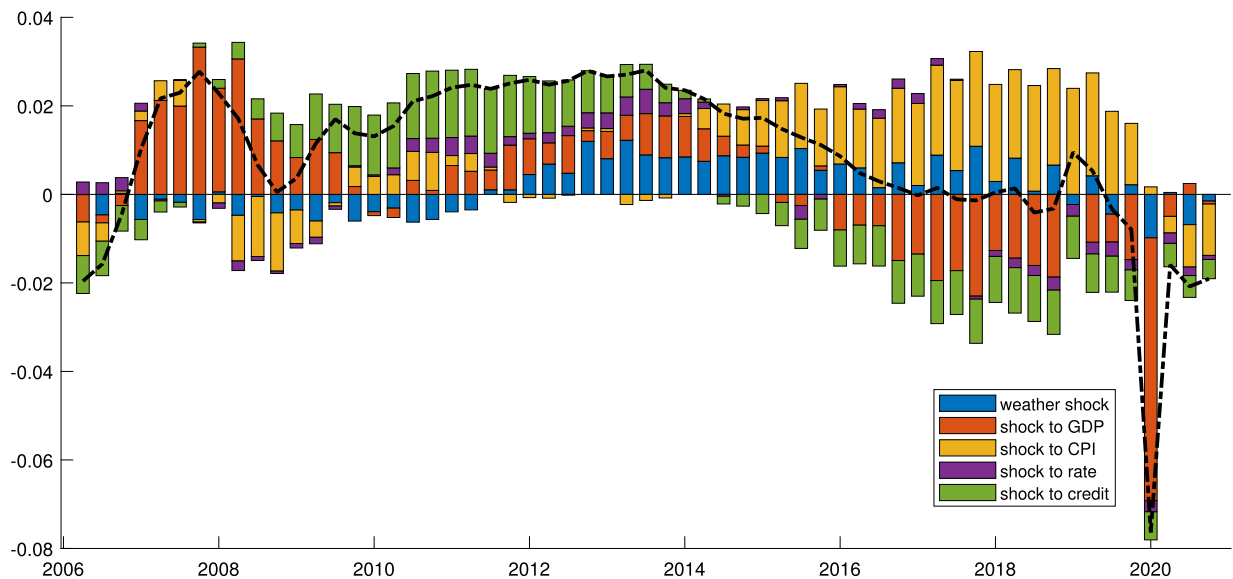


Fig. 6. Historical decomposition for GDP. Source: Authors' calculations. Note: After detrending and demeaning the variables in the SVAR, we decompose the actual deviation of a variable from its trend into contributions from exogenous shocks to the five variables in each period. The dotted black line is the detrended and demeaned GDP series. Following the suggestion of Kilian and Lütkepohl (2017), we discard the first twenty quarters (about 20% of the whole sample) in conducting historical decomposition. This is because the approximation techniques underlying historical decomposition can perform poorly during the initial periods.

Table 1
Variance decomposition.

Variables	EWI	GDP	CPI	Rate	Credit
Q1	100	2.797469	0.001821	0.000553	1.689482
Q4	96.53816	14.59756	1.854872	3.163351	2.187733
Q8	95.21549	18.06606	1.558176	4.368398	2.968091
Q16	94.31764	18.2578	3.386986	4.697204	8.41753
Q200	94.13994	14.20422	11.51271	4.740766	13.81998

Note: Table 1 presents the variance decomposition at the 1st quarter, 4th quarter, the 8th quarter, 16th quarter, and the 200th quarter (which approximates the infinite horizon). The numbers are the percentage of variation of each variable that is explained by EWI shocks.

indicating a substantial lag between the impact and CPI fluctuations. Regarding interest rates and credit obtained by non-financial firms, extreme weather risks can explain 4.7% and 13.8% of their variation in the long term, respectively.

Overall, extreme weather shocks are quantitatively important in explaining fluctuations in the GDP, CPI, and credit obtained by non-financial firms. Extreme weather shocks account for less variation in interest rates, presumably reflecting that the central bank has not regarded climate risk as an important factor in monetary policy until recently. The persistence of the impact of extreme weather shocks on variance decomposition is consistent with IRFs.

Our findings regarding GDP and CPI inflation are comparable to those in the literature. Acevedo et al. (2020) find that, for the median emerging market economy, a 1 °C increase from a temperature of 22 °C lowers growth in the same year by 0.9 percentage points. In comparison, our quarterly SVAR indicates that the cumulative effects of a one-standard-deviation EWI shock on GDP growth in the subsequent year are -0.91%. In terms of inflation, we do not find evidence that the EWI shock significantly affects CPI inflation in the short run. This finding can be reconciled with the findings of Cevik and Jalles (2023), who report ambiguous effects of extreme weather on the inflation rate. To be precise, they find that extreme temperature lowers inflation, but drought and storms raise inflation. Because our EWI index incorporates information on high temperature, low temperature, precipitation, drought, and wind, it is plausible that the aggregation of heterogeneous effects of subcomponents yields an insignificant overall effect of EWI on inflation.

We conducted a counterfactual analysis to examine the role of monetary policy in the transmission of extreme weather shocks in the macroeconomy. We impose a counterfactual restriction that monetary policy does not react to extreme weather shocks by generating a sequence of hypothetical monetary policy shocks that offset the interest rate decrease triggered by extreme weather shocks. This approach has been widely used in the literature for counterfactual analyses (e.g., see Hamilton and Herrera (2004) and Kilian and Lewis (2011)).

Table 2
Summary statistics of firms listed in China.

	Unit	Obs	Mean	Std.Dev.	Min	Max
<i>loan/asset</i>	%	108225	14.277	13.350	0.110	68.195
<i>value-added</i>	millions of 2020 yuan	102706	954.110	2115.453	0.235	17041.234
<i>sales</i>	millions of 2020 yuan	108149	5082.338	12041.194	16.000	99112.555
<i>asset</i>	millions of 2020 yuan	108225	13874.610	31906.125	120.000	276931.875
<i>tangible</i>	%	108224	40.579	21.222	-22.181	91.749
<i>ROA</i>	%	108224	2.735	4.156	-10.475	26.284
<i>long-debt</i>	%	71761	8.938	9.829	0.000	106.996
<i>short-debt</i>	%	102862	13.412	11.084	0.001	269.332
<i>interest</i>	millions of 2020 yuan	107155	230.466	592.204	0.061	5229.057
<i>NonSOE</i>	indicator	108225	0.568	0.495	0	1
<i>age</i>	year	108225	17.189	5.770	1	32

Notes: (1) The variable *loan/asset* is the firm-level bank loans ratio to total assets, *tangible* is the fraction of tangible assets in total assets, *ROA* is the return on asset, *long-debt* is long-term debt to asset ratio, *short-debt* is the short-term debt to asset ratio. Short-term debt is any debt that is due within one year, while long-term debt is any debt that is due after one year. *interest* is the interest expenses, and *NonSOE* is an indicator for companies in which the state is not the controlling shareholder in the previous quarter.

(2) The average exchange rates of Chinese yuan to one US dollar, one UK pound, and one Euro in 2020 were 6.904, 8.857, and 7.878, respectively.

The results indicate whether monetary policy reacts to extreme weather shocks or not, the output level is persistently below its initial level after the shock.⁴ Meanwhile, if the central bank does not react to extreme weather shocks, the drop in output is significantly larger after the 4th quarter. Under the counterfactual, the drop in GDP at the 20th quarter following a one-standard-deviation extreme weather shock is -0.49%, which is deeper than the drop of -0.42% estimated from the actual data. The results suggest that the central bank should adequately consider the effects of extreme weather shocks when implementing monetary policy.

To ensure that our counterfactual analysis does not involve unduly large shocks, we compute the ‘modesty statistic’ of Leeper and Zha (2003) for the implied offsetting shocks that deliver the counterfactual constraint of a constant interest rate. The absolute value of the modesty statistics is well below the critical value of two at all the horizons, meaning that the offsetting shocks are ‘modest’ and their materialization is unlikely to induce agents to adjust their expectation formation and beliefs about the structure of the economy.⁵

4. Firm-level evidence

Our SVAR analysis provides evidence that extreme weather shocks have persistent negative effects on the aggregate output and credit obtained by non-financial firms. A natural question is how the effects of extreme weather can propagate beyond the initial physical damage. In this section, we use firm-level panel regressions to focus on the mechanisms through which weather shocks could affect credit access in the firm sector.

4.1. Firm-level data

We obtained data on companies listed in China between the first quarter of 2006 and fourth quarter of 2020 from the WIND database. We excluded financial companies, companies with the designation ‘ST’ or ‘PT,’ and companies reporting missing values.⁶

Our eventual dataset included an unbalanced panel of 4,186 firms and 108,225 firm-quarter observations. The summary statistics are reported in Table 2. The mean ratio of bank loans to total assets was 14.28%, with a standard deviation of 13.35%. The variable range is 0.11% to 68.20%; thus, the degree of dependence on bank loans varies considerably among the firms. Within the sample, 56.8% of firm-quarter observations are accounted for by non-state-owned enterprises (non-SOEs). Relative to SOEs, non-SOE firms are more likely to be financially constrained, making them more sensitive to extreme weather conditions.

4.2. Firm-level panel regression results

4.2.1. Baseline credit regressions

First, we estimate whether extreme weather affects bank loans obtained by companies, using the following fixed-effects regression model:

$$(loan/asset)_{i,t} = \beta_1 EW I_{t-1} + \beta_2 X_{i,t-1} + d_{year} + d_i + \epsilon_{i,t} \quad (3)$$

⁴ The figure of the counterfactual analysis is available upon request.

⁵ The figure of the test statistics is available upon request.

⁶ The designation ‘ST’ (Special Treatment) and ‘PT’ (Particular Transfer), indicates that a firm has reported a negative profit for two consecutive years and three consecutive years, respectively. These firms face a significant risk of de-listing due to regulatory rules in China.

Table 3
Baseline regressions.

Dep. Variable: <i>loan/asset</i>	(1)	(2) base	(3) jump	(4) jump	(5) w/o 2008-2009	(6) w/o 2020	(7) macro
<i>EWI</i>	-0.220*** (0.040)	-0.232*** (0.030)	-0.374*** (0.092)	-0.586*** (0.103)	-0.140*** (0.032)	-0.214*** (0.031)	-0.247*** (0.032)
<i>tangible</i>		-0.180*** (0.003)	-0.180*** (0.003)	-0.180*** (0.003)	-0.185*** (0.003)	-0.182*** (0.003)	-0.180*** (0.003)
<i>ROA</i>		-0.153*** (0.010)	-0.153*** (0.010)	-0.153*** (0.010)	-0.153*** (0.011)	-0.159*** (0.012)	-0.148*** (0.010)
<i>age</i>		0.181* (0.096)	0.183* (0.096)	0.182* (0.096)	0.212** (0.098)	0.062 (0.113)	0.170* (0.096)
<i>ln(asset)</i>		0.274*** (0.084)	0.279*** (0.084)	0.278*** (0.084)	0.053 (0.090)	0.268*** (0.091)	0.284*** (0.084)
<i>leverage</i>		-0.005 (0.012)	-0.004 (0.012)	-0.013 (0.012)	-0.077*** (0.014)	0.130*** (0.015)	-0.538*** (0.035)
<i>gepu</i>							-0.004* (0.002)
<i>mpu_us</i>							0.006*** (0.002)
<i>vix</i>							-0.052*** (0.009)
<i>m2</i>							0.022 (0.027)
<i>gdp</i>							-0.133*** (0.020)
<i>credittogdp</i>							0.433*** (0.027)
<i>_cons</i>	21.168*** (0.099)	19.993*** (2.519)	19.740*** (2.519)	19.835*** (2.519)	23.834*** (2.689)	22.808*** (2.782)	17.258*** (2.577)
<i>Dependent Variable's Mean</i>	14.277	14.277	14.277	14.277	14.277	14.842	14.842
<i>Controls</i>	N	Y	Y	Y	Y	Y	Y
<i>Year FE</i>	Y	Y	Y	Y	Y	Y	Y
<i>Firm FE</i>	N	Y	Y	Y	Y	Y	Y
<i>N</i>	108225	108225	108225	108225	98723	96087	108225
<i>adj. R²</i>	0.172	0.578	0.578	0.578	0.585	0.586	0.580

Notes: (1) The dependent variable is bank loans ratio to total assets. Columns 3 and 4 use jump indicators to measure significant extreme weather risks as large jumps in the EWI index. Columns 5 and 6 exclude observations from 2008 and 2009, and 2020. (2) *tangible* is the fraction of tangible assets in total assets; *ROA* is firm's return on assets; *age* is firm age; *ln(asset)* is the log of firm-level assets; *leverage* is the aggregate leverage of non-financial firms. *gepu* is the global economic uncertainty index; *mpu_us* is the monetary policy uncertainty for US; *vix* is the VIX index, *m2*, *gdp* and *credittogdp* are monetary supply, GDP growth rate and credit to GDP gap in China, respectively. (3) We use a one-quarter lag of *EWI* in the loan regression; (4) Robust standard errors are in parentheses. (5) All regressions include year fixed effects. Columns 2-7 include firm fixed effects. (6) ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

where i and t are the indices for the firms and quarters, respectively. The dependent variable was the ratio of bank loans to total assets. EWI_t denotes the Chinese climate risk index we constructed. The vector of control variables $X_{i,t}$ includes the log of firm-level assets ($lnasset$), firm age (age), fraction of tangible assets in total assets ($tangible$), and return on assets (ROA). Quantities d_{year} and d_i are the year- and firm-fixed effects in the regression, respectively. Considering that the impact of extreme weather risks on credit most likely occurs with a lag, we use a one-quarter lag of *EWI* in the regression.

Table 3 presents the results of the main regressions. In the first column, we run a simple regression of credit on the *EWI* and year fixed effects. As expected, we find that extreme weather reduces bank loans obtained by companies. An adjusted R^2 value of 0.172 indicates that a significant portion of the variation in loans can be accounted for by the *EWI* and year fixed effects.⁷ Column (2) is our baseline specification, and according to it, a one-unit increase in *EWI* causes the bank loans to assets ratio to drop by 0.232 percentage point. The estimated effect is significant in both statistical and economic terms.

In columns (3) and (4), we use jump indicators for the *EWI* as alternative independent variables to estimate the effect of extreme weather shocks. In column (3) (column (4), respectively), we define a jump event as a situation where the Hodrick-Prescott (HP) filtered *EWI* index (the original *EWI* time series) exceeds the mean value by more than 1.65 times the standard deviation. The *EWI* jump indicator takes the value of one if at least one jump event occurs in a quarter and zero otherwise. The coefficient on the jump indicator in column (3) (column (4), respectively) indicates that if a jump event or events occurs in a quarter, the bank loan to assets ratio will drop by 0.374 (0.586) percentage point.

We exclude observations from 2008 to 2009 from the regression in column (5) to check whether the estimated effect of the *EWI* is confounded by the effects of demand shocks during the global financial crisis. Similarly, in Column (6), we exclude observations from 2020 to ensure that our findings are not driven by the devastating effect of the COVID-19 pandemic. In these two columns, the

⁷ When we drop the year fixed effects from the regression, a bi-variate regression of loans on *EWI* yields an adjusted R^2 of 0.151.

estimated coefficients at -0.140 and -0.214 are somewhat smaller in magnitude than the benchmark of -0.232 but remain significant. To rule out the possibility that our results could be driven by global and national macroeconomic conditions, in column (7), we control for the index of global economic policy uncertainty of Baker et al. (2016), the index of US monetary policy uncertainty of Baker et al. (2016), and China's M2 growth rate, GDP growth rate, and credit gap. Again, the coefficient estimate of -0.247 is very similar to that of the baseline in column (2). Overall, there is strong evidence that EWI shocks reduce companies' bank loans. The results of the robustness exercises that have been conducted are available in the online appendix.

Naturally, one would expect that firms respond mainly to local extreme weather events. To verify this possibility, we also construct the EWI at the provincial level and use it to replace the national EWI in the benchmark regression. The coefficient from the regression, which is not reported here to conserve space, is -0.015 with a standard deviation of 0.015. Thus, the estimated coefficient on provincial EWI is much smaller than the coefficient on the national EWI in the benchmark regression (-0.232) and is statistically insignificant. One rationalization is that because listed firms often own production sites in multiple provinces and sell to the national market or markets beyond their home province, they are more sensitive to extreme weather that has a national impact. Therefore, we will use the national EWI in regressions in the remaining parts of the paper. In the next two subsections, we examine whether the effects of extreme weather shocks affect loans through credit risk and expectation channels.

4.2.2. The credit risk channel

In this section, we examine the roles of the four types of credit risk-related factors in the propagation or moderating effects of extreme weather shocks. First, we study whether extreme weather shocks reduce loans through expected default frequency (EDF), a measure of default risk. We construct the EDF index at the firm-quarter level by following the methods of Bharath and Shumway (2008) and Brogaard et al. (2017)⁸ and regress it on EWI. In a regression of EDF on EWI, year fixed effects and firm fixed effects, a one-standard deviation increase in EWI increases the EDF by 0.003 and the effect is significant at 1% level. When firm-level controls are added, the coefficient on EWI is 0.001 and significant at 10% level.⁹ As default risk is naturally associated with a lower level of bank loans, taken together, the results suggest that extreme weather risks depress bank loans by elevating firm-level default risk.

Second, we analyzed the role of collateral assets. On the one hand, if extreme weather shocks inflict significant damage on fixed assets that can be used as collateral, firms' ability to obtain loans will be hindered. However, a large stock of fixed assets can help a firm retain credit access in the aftermath of extreme weather shocks. To estimate the net influence of collateral, Table 4 sequentially includes the interactions of three proxy variables for collateral assets with EWI. The three proxy variables (*low fixed-assets*, *low non-current-assets* and *low tangible-assets*¹⁰) are indicators of a firm with specified assets below the median level in the previous quarter. The interaction coefficients are all significant and negative, indicating that firms with less collateral face more difficulty in obtaining bank loans after extreme weather shocks. Thus, the availability of collateral assets moderates the effects of extreme weather shocks.

Third, we examine whether macroeconomic credit risk interacts with extreme weather to cause a larger drop in credit financing. As central bank policies may mitigate or amplify the impact of weather shocks, we introduce an indicator for tight monetary policy (*tight MP*) and its interaction with extreme weather. The variable *tight MP* is an indicator variable that takes the value of one for an increase in the quarterly average of the R007 interest rate and zero otherwise. In column (1) of Table 5, the coefficients of EWI remain negative and the interaction between *tight MP* and extreme weather is negative. The negative sign of the interaction term suggests that the adverse effects of extreme weather on firm-level loans are more severe when monetary policy is tighter. Intuitively, when the economy is hit by extreme weather in a tight monetary environment, firms face additional challenges in financing and maintaining their level of output.

In addition to monetary policy, we introduce three variables that capture different aspects of aggregate credit risk: the economic policy uncertainty (EPU) index of Baker et al. (2016), financial stress index of Park and Mercado (2014), and aggregate leverage of non-financial firms (defined as the ratio of non-financial firms' total debt to GDP). EPU is an important indicator of credit risk, as it is associated with higher interest rates on bank loans and, hence, borrowers' credit risk (Ashraf and Shen, 2019). The financial stress measure proposed by Illing and Liu (2006) and adopted by Park and Mercado (2014) is a commonly used gauge of financial stress, and it is found to be positively related to expected financial loss, risk, or uncertainty. Hennessy and Whited (2007) make the case that the leverage ratio provides a measure of financial friction because the leverage ratio provides information about bankruptcy costs.¹¹

Columns (2) through (4) in Table 5 report the regressions with economic policy uncertainty, financial stress, and leverage. The coefficients of the interaction terms are all negative and statistically significant, except for in the fourth column. Therefore, extreme weather, in general, has a more adverse effect on bank loans obtained by firms when economic policy uncertainty levels and financial stress are high.

⁸ The expected default frequency is obtained using the KMV model offered through Moody's Investors Service, which is based on Merton (1974)'s distance to default (DD) model. The KMV model defines the expected default frequency as a function of "distance to default", i.e. the difference between the expected asset value at the analysis horizon and the "default point" normalized by the standard deviation of asset returns. This default point equals the sum of all short term debt and half the long term debt.

⁹ The regression results which are not reported here to conserve space, are available upon request.

¹⁰ We compute tangible-assets as the difference between total assets and intangible assets.

¹¹ While Hennessy and Whited (2007) focuses on leverage at the firm level, we use its aggregate counterpart, because the aggregate leverage ratio likely indicates credit risk at the firm level.

Table 4
Credit risk channel: the role of collateral.

Dep. Variable: <i>loan/asset</i>	(1) fixed assets	(2) non-current assets	(3) tangible assets
<i>EWI</i> ² <i>low fixed assets</i>	-0.438*** (0.043)		
<i>EWI</i> ² <i>low non-current assets</i>		-0.292*** (0.041)	
<i>EWI</i> ² <i>low tangible assets</i>			-0.113*** (0.040)
<i>EWI</i>	-0.288*** (0.031)	-0.257*** (0.030)	-0.242*** (0.029)
<i>low fixed assets</i>	-0.327*** (0.108)		
<i>low non-current assets</i>		0.244** (0.108)	
<i>low tangible assets</i>			0.039 (0.110)
<i>Dependent Variable's Mean</i>	14.277	14.277	14.277
<i>Controls</i>	Y	Y	Y
<i>Year FE</i>	Y	Y	Y
<i>Firm FE</i>	Y	Y	Y
<i>N</i>	104627	105574	105509
<i>adj. R²</i>	0.578	0.579	0.592

Notes: (1) The dependent variable is bank loans ratio to total assets. (2) *low fixed assets* is an indicator for a firm with fixed assets below the median level in the previous quarter; *low non-current assets* is an indicator for a firm with non-current assets below the median level in the previous quarter; *low tangible assets* is an indicator for a firm with tangible assets below the median level in the previous quarter. (3) We use a one-quarter lag of *EWI* in the regression. (4) The numbers in parentheses are robust standard errors. (5) All regressions include control variables as in column 2 of Table 3, year fixed effects, and firm fixed effects. (6) ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

Finally, we explore the cross-province variation in local government debt to provide further evidence for the credit risk channel. Our premise is that if the credit environment is more fragile in provinces where local government borrowing crowds out corporate borrowing, the drop in loans obtained by listed companies following an extreme weather shock should be larger in magnitude. To test this conjecture, we include in the regression the per capita value of outstanding bonds issued by all financing vehicles owned by all levels of local government in a province (*LGFV bond*) and its interaction with *EWI*.¹² Column (6) shows that the interaction term is negative and significant. Overall, we can conclude that the drop in loans is greater when macroeconomic credit risk is higher.

4.2.3. The expectation channel

Besides interacting with credit risk, extreme weather can reduce loans obtained by listed companies, by dampening the expectations of the financial markets and firms. First, when extreme weather disrupts a firm's normal operations, analysts' and investors' expectations become more pessimistic. To directly test the expectation channel, we regress four measures of firm-level ratings by stock analysts (obtained from the CSMAR database), on the *EWI*. We code the rating opinions of 'sell', 'reduce', 'neutral', 'increase', and 'buy' as 1, 2, 3, 4 and 5 in a categorical variable. With this definition, an increase in the rating score indicates an improvement in rating. The specific rating-related measures used in regressions are the mean rating (average ratings based on the opinions of 'sell', 'reduce', 'neutral', 'increase', and 'buy'), the modal rating, a discrete measure of average rating (defined as the rating in the five categories that is the closest to mean rating score), and mean analysts' forecast of net profit. In the first three regressions in Table 6, the estimated coefficients are all negative and statistically significant. In column (4), the results show extreme weather shocks significantly decreasing analysts' forecasts of firm profits.¹³ These findings provide evidence that, extreme weather dampens market expectations, in turn hindering the credit access of listed companies.

For further analysis, we run regressions with interactions between the *EWI* and several expectation-related macroeconomic variables. They are indicators of the Entrepreneurs' Confidence Index (ECI), the Purchasing Managers' Index (PMI), and the new-order component of the PMI index.¹⁴ Our conjecture is that, when the corporate sector is more confident, the negative effect of extreme

¹² Up to 2015, local governments were prohibited from issued government bonds in the financial market. To circumvent this restrictions, local governments use financing vehicles (LGFVs) to raise funds from the corporate bonds market. Throughout the sample, the values of outstanding bonds by LGFVs provide a consistent measure of implicit indebtedness of governments at the provincial level.

¹³ When we estimate the regressions of the modal rating and the measure of discrete average rating with a Ordinal Logit regression, the effects of *EWI* remain negative and significant. We present the results from linear regressions in Table 6 to facilitate a quick and approximate comparison.

¹⁴ These three variables are from National Bureau of Statistics.

Table 5
Credit risk channel: macroeconomic risk.

Dep. Variable: <i>loan/asset</i>	(1)	(2)	(3)	(4)	(5)
<i>EWI*tight MP</i>	-0.371*** (0.046)				
<i>EWI*EPU</i>		-0.206*** (0.029)			
<i>EWI*stress</i>			-0.289*** (0.083)		
<i>EWI*leverage</i>				-0.006 (0.005)	
<i>EWI*LGFV bond</i>					-0.107*** (0.029)
<i>EWI</i>	-0.206*** (0.029)	-0.099** (0.039)	-0.238*** (0.030)	-0.248*** (0.033)	-0.199*** (0.032)
<i>tight MP</i>	-0.101 (0.063)				
<i>EPU</i>		-0.243** (0.118)			
<i>stress</i>			0.528*** (0.119)		
<i>leverage</i>				-0.006 (0.012)	
<i>LGFV bond</i>					0.209*** (0.063)
<i>Dependent Variable's Mean</i>	14.277	14.277	14.277	14.277	14.277
<i>Controls</i>	Y	Y	Y	Y	Y
<i>Year FE</i>	Y	Y	Y	Y	Y
<i>Firm FE</i>	Y	Y	Y	Y	Y
<i>N</i>	108225	108225	108225	108225	92810
<i>adj. R²</i>	0.579	0.579	0.579	0.578	0.598

Notes: (1) The dependent variables is bank loans ratio to total assets. (2) *tight MP* is an indicator variable for an increase in the quarterly average of the R007 interest rate; *EPU* is the Economic Policy Uncertainty index from Baker et al. (2016); *stress* is the financial stress index from the Park and Mercado (2014); *leverage* is the aggregate leverage of non-financial firms; *LGFV bond* is the per capita value of outstanding bonds issued by all financing vehicles owned by all levels of local government in the province where a firm located. (3) We use a one-quarter lag of *EWI* in the regression. (4) Robust standard errors are in parentheses. (5) All regressions include control variables as in column 2 of Table 3, year fixed effects, and firm-fixed effects. (6) ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 6
Expectation channel: rating.

Dep. Variable:	(1) avg-rate	(2) mode-rating	(3) discrete-avg-rating	(4) profit-forecast
<i>EWI</i>	-0.008*** (0.002)	-0.008*** (0.002)	-0.010*** (0.002)	-0.010*** (0.003)
<i>Dependent Variable's Mean</i>	4.273	4.370	4.237	19.708
<i>Controls</i>	Y	Y	Y	Y
<i>Year FE</i>	Y	Y	Y	Y
<i>Firm FE</i>	Y	Y	Y	Y
<i>N</i>	57356	57356	57356	36106
<i>adj. R²</i>	0.330	0.283	0.303	0.863

Notes: (1) The dependent variables are the mean rating (average ratings based on the opinions of 'sell', 'reduce', 'neutral', 'increase', and 'buy'), the modal rating, a discrete measure of average rating (defined as the rating in the five categories that is the closest to mean rating score), and mean analysts' forecast of net profit. A higher rating score indicates a better rating. (2) Robust standard errors are in parentheses. (3) All regressions include control variables as in column 2 of Table 3, year fixed effects, and firm-fixed effects. (4) ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

weather on loans is likely to be mitigated. Table 7 shows that all interaction terms of confidence-related variables have positive and significant coefficients. Thus, higher corporate confidence moderates negative effects of extreme weather shocks. Overall, both sets of regressions provide strong evidence for the expectation channel.

Table 7
Expectation channel: confidence.

Dep. Variable: <i>loan/asset</i>	(1)	(2)	(3)
<i>EWI*Confidence</i>	0.019*** (0.003)		
<i>EWI*PMI-New-order</i>		0.042*** (0.009)	
<i>EWI*PMI</i>			0.073*** (0.014)
<i>EWI</i>	-0.271*** (0.031)	-0.208*** (0.033)	-0.196*** (0.034)
<i>Confidence</i>	-0.045*** (0.006)		
<i>PMI-New</i>		-0.108*** (0.016)	
<i>PMI</i>			-0.156*** (0.027)
<i>Dependent Variable's Mean</i>	14.277	14.277	14.277
<i>Controls</i>	Y	Y	Y
<i>Year FE</i>	Y	Y	Y
<i>Firm FE</i>	Y	Y	Y
<i>N</i>	108225	108225	108225
<i>adj. R²</i>	0.579	0.579	0.579

Notes: (1) The dependent variables is bank loans ratio to total assets. (2) *Confidence* is the entrepreneurs' confidence; *PMI-New-order* is the purchasing managers' index: new order; *PMI* is the overall purchasing managers' index. (3) We use a one-quarter lag of *EWI* in the regression. (4) Robust standard errors are in parentheses. (5) All regressions include control variables as in column 2 of Table 3, year fixed effects, and firm-fixed effects. (6) ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

4.3. Heterogeneity

4.3.1. Firm-level heterogeneity

Because the impact of EWI shocks may be more severe on underprivileged or more fragile firms, especially in a developing country such as China, we examine three types of firm heterogeneity: ownership, firm age, and firm size. We include the interactions of four variables that measure heterogeneity with EWI in the regressions, and report the results in Table 8. In the first column, the variable *NonSOE* identifies companies for which the state is not the controlling shareholder in the previous quarter. The interactive coefficient is negative, indicating a higher drop in non-state-owned firm bank loans, compared to that obtained by State-Owned Enterprises (SOEs). This finding confirms that, SOEs maintain better access to financing after occurrence of extreme weather shocks.

In the second column, the variable *young* identifies firms aged below the median level in the previous quarter. The negative and significant coefficient of the interaction between EWI and the *young* indicator indicates that the credit financing of younger firms are reduced to a greater extent following an extreme weather shock. The variable *small-size* (*small-sales*, respectively) is an indicator of a firm with asset (sales) below the median level in the previous quarter. In Columns (3) and (4), coefficients of the interactions of these two variables with EWI are both negative and significant. Therefore, smaller firms' credit financing is more vulnerable to extreme weather shocks.

4.3.2. Sector-level heterogeneity

While much of the climate-impact literature focus on agriculture and manufacturing, extreme weather shocks likely affect a wide range of economic activities. Therefore, we run separate regressions for loans obtained by listed firms, in three main economic sectors: primary (agricultural), secondary (industrial), and tertiary (services). In Table 9, the coefficients on EWI are all negative and similar in magnitude, albeit for the subsample of firms in the primary sector, EWI has an insignificant coefficient, with a t-statistic of -1.13. We believe that the lack of statistical significance is likely caused by the relatively small sample size. In columns (4)–(6), the interaction terms between EWI and *log non-current asset* are also added. The sign, size, and significance level of the EWI coefficients, are comparable to those in columns (1)–(3). Meanwhile, the interaction term between the EWI and *non-current asset*, is positive and significant in the secondary sector. These results suggest that, the impact of extreme weather on loans is less severe for firms in the secondary sector which have more non-current assets.

4.4. Value-added regression

Our analysis of macroeconomic data suggests that extreme weather shocks have significant and negative effects on GDP. In this subsection, we examine whether there is a similar drop in firm-level output. Thus, we use value added as the dependent variable and estimate the following model:

Table 8
Firm-level heterogeneity.

Dep. Variable: <i>loan/asset</i>	(1)	(2)	(3)	(4)
<i>EWI*NonSOE</i>	-0.345*** (0.038)			
<i>EWI*young</i>		-0.248*** (0.042)		
<i>EWI*low-fixed assets</i>			-0.442*** (0.039)	
<i>EWI*low-sales</i>				-0.417*** (0.039)
<i>EWI</i>	-0.273*** (0.029)	-0.271*** (0.031)	-0.279*** (0.029)	-0.285*** (0.029)
<i>low-fixed assets</i>			-0.323*** (0.121)	
<i>low-sales</i>				-0.429*** (0.114)
<i>Dependent Variable's Mean</i>	14.277	14.277	14.277	14.842
<i>Controls</i>	Y	Y	Y	Y
<i>Year FE</i>	Y	Y	Y	Y
<i>Firm FE</i>	Y	Y	Y	Y
<i>N</i>	108225	99138	106713	106974
<i>adj. R²</i>	0.579	0.581	0.580	0.580

Notes: (1) The dependent variable is bank loans ratio to total assets. (2) *NonSOE* is an indicator for companies in which the state is not the controlling shareholder in the previous quarter; *young* is an indicator for a firm with age below the median level in the previous quarter; *low-fixed assets* is an indicator for a firm with fixed assets below the median level in the previous quarter. *low-sales* is an indicator for a firm with sales below the median level in the previous quarter. (3) We use a one-quarter lag of *EWI* in the regression. (4) The numbers in parentheses are robust standard errors. (5) All regressions include control variables as in column 2 of Table 3, year fixed effects, and firm-fixed effects. (6) ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

Table 9
Industry-level heterogeneity.

Dep. Variable: <i>loan/asset</i>	(1) primary	(2) secondary	(3) tertiary	(4) primary	(5) secondary	(6) tertiary
<i>EWI</i>	-0.222 (0.197)	-0.241*** (0.034)	-0.216*** (0.063)	-0.242 (0.207)	-0.315*** (0.035)	-0.233*** (0.067)
<i>EWI*low non-current asset</i>				0.056 (0.110)	0.143*** (0.015)	0.039 (0.026)
<i>Dependent Variable's Mean</i>	18.552	14.287	13.714	18.552	14.287	13.714
<i>Controls</i>	Y	Y	Y	Y	Y	Y
<i>Year FE</i>	Y	Y	Y	Y	Y	Y
<i>Firm FE</i>	Y	Y	Y	Y	Y	Y
<i>N</i>	3033	80822	24348	2984	78840	23888
<i>adj. R²</i>	0.594	0.593	0.529	0.595	0.594	0.533

Notes: (1) The dependent variable is bank loans ratio to total assets. (2) *low non-current asset* is an indicator for a firm with non-current assets below the median level in the previous quarter; Column 1 and 4 are sub-samples of primary sector; Column 2 and 5 are sub-samples of secondary sector; Column 3 and 6 are sub-samples of tertiary sector. (3) We use the one-quarter lag of *EWI* in the regression. (4) Robust standard errors are in parentheses. (5) All regressions include control variables as in column 2 of Table 3, year fixed effects, and firm-fixed effects. (6) ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

$$\ln(\text{value added}_{i,t}) = \gamma_1 EWI_{i,t-2} + \gamma_2 X_{i,t-1} + d_{\text{year}}^{VA} + d_i^{VA} + \epsilon_{i,t-2} \quad (4)$$

where i and t are the indices for the firms and quarters, respectively. The dependent variable was log value added. The vector of the control variables is the same as in the baseline regression for loans. Quantities d_{year}^{VA} and d_i^{VA} are the year fixed effects and firm fixed effects in the value-added regression, respectively. We use the two-quarter lag of *EWI* in the value added regression to achieve a balance between model's fit and the sample size. We find that using the two-quarter lag of *EWI* improves the model's fit compared with one-quarter lag, as it generates a higher adjusted R-squared and smaller AIC and BIC. The improvement of model's fit is marginal if we choose three or four quarter lag, while the higher order lags would result in a smaller sample size.

In the absence of value-added reported by listed companies in their financial statements, we construct value-added by applying the value-added method in GDP accounting of China to information from the balance sheet and income statements. The value added of a firm is calculated as the sum of employee compensation, business taxes and surcharges, depreciation of fixed assets, and operating

Table 10
Value-added regressions.

Dep. Variable: $\ln(VA)$	(1)	(2) baseline	(3) jump	(4) jump	(5) w/o 2008-2009	(6) w/o 2020	(7) macro
<i>EWI</i>	-0.008*** (0.002)	-0.006*** (0.002)	-0.029*** (0.006)	-0.025*** (0.006)	-0.007*** (0.002)	-0.002 (0.002)	0.217*** (0.028)
<i>tangible</i>		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	-0.181*** (0.003)
<i>ROA</i>		0.052*** (0.001)	0.052*** (0.001)	0.052*** (0.001)	0.051*** (0.001)	0.052*** (0.001)	-0.146*** (0.010)
<i>age</i>		0.018*** (0.005)	0.018*** (0.005)	0.018*** (0.005)	0.019*** (0.005)	0.019*** (0.006)	0.186* (0.096)
$\ln(asset)$		0.695*** (0.007)	0.695*** (0.007)	0.695*** (0.007)	0.688*** (0.008)	0.685*** (0.008)	0.289*** (0.085)
<i>leverage</i>		0.001* (0.001)	0.001* (0.001)	0.001 (0.001)	0.001 (0.001)	-0.002** (0.001)	-0.521*** (0.036)
<i>gepu</i>							-0.002 (0.002)
<i>us_mpu</i>							-0.000 (0.002)
<i>vix</i>							-0.032*** (0.009)
<i>m2</i>							0.056** (0.027)
<i>gdp</i>							-0.149*** (0.021)
<i>credittogdp</i>							0.440*** (0.027)
_cons	19.686*** (0.004)	3.712*** (0.183)	3.710*** (0.183)	3.712*** (0.183)	3.862*** (0.196)	3.898*** (0.203)	16.041*** (2.603)
<i>Dependent Variable's Mean</i>	19.375	19.375	19.375	19.375	19.375	19.375	19.375
<i>Controls</i>	N	Y	Y	Y	Y	Y	Y
<i>Year FE</i>	Y	Y	Y	Y	Y	Y	Y
<i>Firm FE</i>	N	Y	Y	Y	Y	Y	Y
<i>N</i>	101766	101766	101766	101766	92734	90343	107267
<i>adj. R²</i>	0.839	0.890	0.890	0.890	0.893	0.891	0.582

Notes: (1) The dependent variable is log value added. Columns 3 and 4 use jump indicators to measure significant weather risks as large jumps in the EWI index. Columns 5 and 6 exclude observations from 2008 and 2009, and 2020. (2) *tangible* is the fraction of tangible assets in total assets; *ROA* is firm's return on assets; *age* is firm age; $\ln(asset)$ is the log of firm-level assets; *leverage* is the aggregate leverage of non-financial firms. *gepu* is the global economic uncertainty index; *us_mpu* is the monetary policy uncertainty for US; *m2*, *gdp* and *credittogdp* are monetary supply, GDP growth rate and credit to GDP gap in China, respectively. (3) We use a two-quarter lag of *EWI* in the loan regression; (4) Robust standard errors are in parentheses. (5) All regressions include year fixed effects. Columns 2-7 include firm fixed effects. (6) ***, ** and * denote significance at the 1%, 5%, and 10% levels, respectively.

profits. Employee compensation and operating profits are readily available from quarterly financial statements, but business taxes, surcharges, and fixed assets depreciation must be imputed. We apply the prevailing sales tax rate to the sales volume, wherever applicable, to obtain the estimated business taxes and surcharges.¹⁵ We set the depreciation of fixed assets equal to a quarter of the reported fixed asset depreciation in the annual statement.

A comparison of the loan and value-added regressions, shows that extreme weather shocks affect credit with a one-quarter lag, and value-added with a two-quarter lag. This indicates that, the response in the credit market is faster than that in the output market. While it is plausible that the drop in credit propagates the impact of the extreme weather shock on output, exploring the dynamics of loans and output at the firm level is beyond the scope of the current study.

Table 10 reports the value-added regression results, which mirror the loan regressions in Table 3. Columns (1) and (2) show that whether we include the control variables or not, the coefficients of EWI are negative, statistically significant, and similar in magnitude. According to column (2), which is our benchmark for value-added regression, a one-unit change in the EWI index reduces a firm's value-added by 0.8% ($-0.008 \cdot 100\%$). In columns (3) and (4), when we use jump indicators to replace the EWI, similar results are obtained. In columns (5) (respectively, column (6)), we exclude observations from 2008 to 2009 (from 2020), and the coefficient on EWI remains negative, albeit not significant in column (6). In column (6), we control for a set of global and national macroeconomic variables and obtain similar results, indicating that other macroeconomic events are unlikely to drive our benchmark value-added results.

¹⁵ The sales tax was applied to nine types of industries and economic activities before 2016. These industries and activities are transportation (3% tax rate), construction (3%), telecommunication (3%), culture and sports (3%), finance and insurance (5%), services (5%), transfer of intangible assets (5%), sales of real estate (5%), and entertainment (5%–20%). In 2016, the sales tax was replaced by a value-added tax. Hence, we do not include the imputed sales tax in the value-added for firm observations, in and after 2016.

5. Conclusion

Climate change has had significant economic consequences. An important impact pathway is the damage inflicted by extreme weather events. China has a large land area that frequently experience extreme weather events. This study constructs an EWI to measure Chinese extreme weather risk between 2001 and 2020, and estimates its effect on China's macroeconomy and firms listed in China.

We first estimate a SVAR model of EWI, GDP, CPI, interest rate, and total credit obtained by non-financial firms from 2001 to 2020. The main finding is that an adverse extreme weather shock depresses GDP and credit in the long term, and lowers the interest rate in the short term. Following a one standard deviation shock to EWI index, GDP drops by 0.5% in the third quarter. The variance decomposition suggests that extreme weather shocks account for 14.2% of GDP fluctuations in the long run. In a counterfactual analysis, we find that the impact of a extreme weather shock becomes more severe if monetary policy does not loosen after the extreme weather shocks.

Using a panel dataset of firms listed in China from 2006 to 2020, we then estimate the effects of extreme weather on firms' credit financing. Similar to our macroeconomic analysis, we find that loans drop after extreme weather events. We find evidence of extreme weather shocks impacting credit financing, through the credit risk and the expectation channels. The effect of extreme weather shocks is more pronounced when firm-level and macro-level credit risks are higher. Regarding the expectation channel, we find that extreme weather shocks dampen market expectations, and their impact on firms is larger when firm managers' expectations and confidence are low. Taken together, our macroeconomic and microeconomic analyses provide strong evidence that extreme weather has a significant impact on the Chinese economy, particularly on the credit access of Chinese firms.

This study explores the infrequently researched area of the effects of extreme weather events on the Chinese economy. There is considerable future scope of research, to quantify the economic consequences of climate change in China and other countries. Our future intention is to use additional Chinese micro and regional data, for further similar studies. It would also be worthwhile, to study the process of monetary and credit policy usage, by central banks and financial authorities, to mitigate the effects of extreme weather events.

Declaration of competing interest

The authors declare that they have no relevant or material financial interests that relate to the research described in this paper.

Data availability

The authors will make the data and codes available publicly via zenodo.org at <https://doi.org/10.5281/zenodo.10912478>.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GPT4.0 developed by OpenAI in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jebo.2024.03.025>.

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